CCAFusion: Cross-Modal Coordinate Attention Network for Infrared and Visible Image Fusion

Xiaoling Li, Yanfeng Li, Houjin Chen, Yahui Peng, and Pan Pan

***Abstract*— Infrared and visible image fusion aims to generate one image with comprehensive information. It can maintain rich texture characteristics and thermal information. However, for existing image fusion methods, the fused images either sacrifice the salience of thermal targets and the richness of textures or introduce the interference of useless information like artifacts. To alleviate these problems, an effective cross-modal coordinate attention network for infrared and visible image fusion called CCAFusion is proposed in this paper. To fully integrate complementary features, the cross-modal image fusion strategy based on coordinate attention is designed, which consists of the feature-awareness fusion module and the feature-enhancement fusion module. Moreover, a multiscale skip connection-based network is employed to obtain multiscale features in the infrared image and the visible image, which can fully utilize the multi-level information in the fusion process. To reduce the discrepancy between the fused image and the input images, a multiple constrained loss function including the base loss and the auxiliary loss is developed to adjust the gray-level distribution and ensure the harmonious coexistence of structure and intensity in fused images, thereby preventing the pollution of useless information like artifacts. Extensive experiments conducted on widely used datasets demonstrate that our CCAFusion achieves superior performance over state-of-the-art image fusion methods in both qualitative evaluation and quantitative measurement. Furthermore, the application to salient object detection reveals the potential of our CCAFusion for high-level vision tasks, which can effectively boost the detection performance.**

***Index Terms*— Infrared and visible image fusion, attention mechanism, cross-modal fusion strategy, coordinate attention, multiple constrained loss function.**

I. INTRODUCTION

INFRARED and visible image fusion aims to generate one image with comprehensive information. It can maintain rich texture characteristics in the visible image and thermal information in the infrared image [1], [2], [3], so as to promote the ability of crucial information acquisition in challenging circumstances, such as light change, occlusion, and background clutter. Image fusion has been a vital preprocessing for many applications to accurately identify the underlying phenomena in complex scenes. Especially, image fusion can be beneficial in a broad range of multimodal high-level vision tasks, such as video surveillance [4], semantic segmentation [5], and object detection [6], [7].

The early research on image fusion mostly adopts multi-scale transform approaches, sparse representation approaches, saliency-based approaches, subspace-based approaches, hybrid approaches, and other approaches to merge different modal images [8], [9]. These traditional fusion approaches have a positive impact on the infrared and visible image fusion task. However, the hand-crafted features they adopted are with limited representation ability, resulting in spatial distortion in the fused image, such as artifacts and incomplete information.

Recently, deep learning-based methods have been introduced into image fusion tasks due to their excellent ability in extracting features. Existing deep learning-based fusion approaches are mainly based on the convolutional neural network (CNN) framework [10], [11], the generative adversarial network (GAN) framework [12], [13], [14], and the auto-encoder (AE) framework [15], [16]. Although the deep learning-based image fusion methods generate promising fused results, some limitations still exist. First, some image fusion methods employ straightforward ways to transfer features from the source images to the fused images. For instance, the predesigned fusion rules in [17] and [18] simply averaged the feature maps to obtain the fused features. However, the extracted features from the source images may be incompletely utilized in this direct way. Second, it is difficult to balance the discrepancies between the source images and the fused image due to the lack of desirable ground-truth data in image fusion, even introducing the interference of useless information like artifacts in the fused image. To this end, some works explored image fusion methods based on the GAN framework [19], [20]. Although most of the details were preserved by the GAN-based fusion methods, fully exploiting the representative information in the multimodal images is difficult for the GAN network. Third, the features extracted by AE-based fusion methods were merged without any appropriate guidance when integrating the meaningful information of the source images [21], [22], [23]. To distinguish the importance of different features, the attention mechanism is introduced into the AE-based fusion frameworks recently. For instance, a fusion strategy based on the convolutional block attention module (CBAM) [24] was put forward to express the significance of each spatial position and each channel [25].

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This image fusion method generated a favorable fused result. However, this fusion method does not explore the cross feature and long-range dependencies in the feature fusion process. Besides, the large-scale pooling is applied to capture local correlations, resulting in the loss of position information.

Motivated by the recent works on the attention mechanism [28] and Densefuse [21], we propose an effective cross-modal coordinate attention network for infrared and visible image fusion called CCAFusion, which aims to alleviate the aforementioned problems in image fusion methods. In contrast to conventional deep learning-based fusion methods, our CCAFusion develops the cross-modal coordinate attention to guide feature fusion by introducing the channel and location importance. Meanwhile, the cross feature and the long-range dependencies are explored in the feature fusion process, so as to avoid the loss of position information caused by the large-scale global pooling. To integrate complementary features, we design the feature-awareness fusion module and the feature-enhancement fusion module in the fusion strategy. In addition, a multiscale skip connection-based network is employed to obtain multiscale features in the infrared image and the visible image. Formally, CCAFusion includes encoder blocks, fusion blocks, and decoder blocks. We utilize the cross-modal coordinate attention fusion blocks to achieve cross-modal fusion at each level. These cross-modal fused features are fed into the decoder blocks based on the skip connection architecture, and the final fused image is gradually reconstructed from the high-level information to the low-level information. Furthermore, aiming at the problem of lacking desirable ground-truth data in image fusion, we design a new loss term to constrain the network training, namely a multiple constrained loss function. To intuitively demonstrate the advantages of CCAFusion, Fig. 1 displays the schematic illustration of infrared and visible image fusion and shows that CCAFusion is effective in maintaining thermal targets and texture features without artifacts.

In summary, our main contributions are described as follows.

1) We propose a novel cross-modal coordinate attention network for infrared and visible image fusion (CCAFusion), which achieves superior performance in the image fusion task. Furthermore, the extended application on salient object detection reveals that CCAFusion can facilitate the detection performance for subsequent high-level vision tasks.
2) We design a simple yet effective cross-modal image fusion strategy that includes the feature-awareness fusion module (FAF) and the feature-enhancement fusion module (FEF). FAF is devised to integrate complementary features based on the single-modal attentional weights. FEF is utilized to promote the complementary features by the cross-modal attentional weights. The ablation experiments on fusion module evaluate the effectiveness of our fusion strategy and indicate that FAF and FEF work in tandem to enable our fusion network to retain abundant information.
3) We develop a multiple constrained loss function for image fusion. Specifically, a new base loss is proposed in this work to impose similar gray-level distributions between the fused image and the source images by the KL divergence, preserving the base information. Besides, the auxiliary loss can maintain the detailed information and the thermal radiation information. CCAFusion trained by the developed loss function can reasonably adjust the gray-level distribution and ensure the harmonious coexistence of structure and intensity, thereby preventing artifact pollution. The ablation experiments on loss function demonstrate the effectiveness and diverse contributions of each loss term.

In the rest of this paper, we review related works of image fusion methods in Section II. In Section III, we describe CCAFusion network in detail. In Section IV, we carry out image fusion experiments on public datasets. In Section V, conclusion is given.

II. Related Works

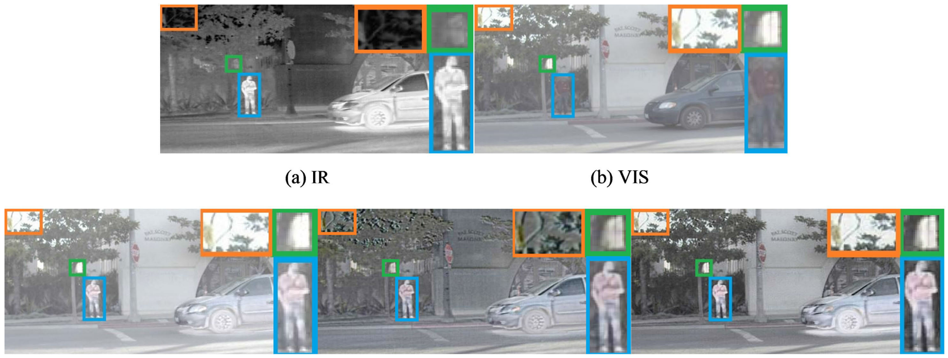
In this section, we first review the classical image fusion algorithms. Then, we introduce recent works about the deep learning-based image fusion methods.

A. Classical Image Fusion Methods

Classical image fusion methods can be divided into six types: multiscale transform approaches, sparse representation approaches, saliency-based approaches, subspace-based approaches, hybrid approaches, and other approaches [8], [9]. The most frequently used methods are multiscale transform and sparse representation.

As a representative branch in classical image fusion methods, multiscale transform approaches aim to decompose the source images into different scales. Subsequently, these features at different scales are combined by an inverse transformation [29]. The conventional multiscale transform approaches include Laplace pyramid transform (LAP) [30], non-subsampled contourlet transform (NSCT) [31], and discrete wavelet transform (DWT) [32]. Madheswari and Venkateswaran [33] suggested an image fusion framework using DWT and combined the thermal image with the visible image using a particle swarm optimization technique, which outperforms most of multi-resolution fusion techniques.

Another typical branch in classical image fusion methods is the sparse representation approach. The goal of sparse representation-based fusion methods is to represent the source images using an over-complete dictionary. The sparse representation coefficients in the over-complete dictionary are then integrated according to fusion rules to obtain the fused image [34]. To gather sufficient feature representation in the dictionary construction, Wang et al. [35] put forward an image fusion method based on morphology, random coordinate encoding, and synchronous orthogonal matching tracking. Considering that the artifact is commonly introduced due to the sliding window technique of the sparse representation, a convolutional sparse representation [36] was developed to learn the coefficient corresponding to the entire source image. Liu et al. [37] adopted the convolutional sparse representation to alleviate the problem of poor detail preservation in image fusion tasks.



(a) IR (b) VIS

(c) PIAFusion (d) U2Fusion (e) CCAFusion (ours)

Fig. 1. Schematic illustration of infrared and visible image fusion. From top left to bottom right: the infrared image (IR), the visible image (VIS), and the fused images of PIAFusion [26], U2Fusion [27], and CCAFusion. Note that CCAFusion preserves clearer thermal targets and more details without artifacts.

B. Deep Learning-Based Image Fusion Methods

Recently, deep learning-based methods [38], [39], [40] have made significant advancements in image fusion due to their excellent ability in extracting features. According to the characteristic of the utilized network architecture in image fusion tasks, the deep learning-based fusion methods can be divided into the CNN-based fusion framework, the GAN-based fusion framework, and the AE-based fusion framework [41].

The CNN-based fusion methods employ convolution operator to extract prominent features and obtain weighting maps. Liu et al. [11] first employed CNN for image fusion. Since this work relied on binary maps, this network was just suitable for multifocus image fusion. To fuse the infrared image and the visible image, Li et al. [42] divided the source images into base parts and detailed parts. For the detailed parts, they created weights based on the features collected by a deep learning network. Afterwards, Zhang and Ma [43] constructed a squeeze-and-decomposition model for fusion tasks, which adopted an adaptive decision block to promote the optimization target and adjusted the weight to change the information proportion from the different input images.

The GAN-based fusion methods make full use of the ability of probability distribution estimation to generate the fused image, which is based on a game between the generator and discriminator. The GAN framework was inventively incorporated into the image fusion task by Ma et al. [20]. They employed a discriminator to drive the generator to obtain the fused image. However, a single discriminator may result in mode collapse, meaning that the fused image favors either the visible image or the infrared image. In addition, FusionGAN in [20] is relatively unstable since it depends on adversarial learning to learn features. As a result, the fused results lack textures in FusionGAN. To preserve the balance between the infrared image and the visible image, Ma et al. [44] suggested a dual-discriminator conditional generative adversarial network. To better integrate the texture information of the source images, Yang et al. [45] constructed an image fusion network based on a texture conditional generative adversarial network.

The AE-based fusion methods adopt an auto-encoder architecture to achieve feature extraction and feature reconstruction. Li and Wu [21] presented an infrared and visible image fusion method based on encoder and decoder, termed as Densefuse. However, Densefuse employed a direct addition strategy in the fusion layer to combine salient features, limiting the fusion performance. To alleviate this problem, Xu et al. [46] put forward an AE-based image fusion framework with a pixel-wise classification saliency-based fusion rule, which broke through the bottleneck of applying deep learning to fusion rules. More recently, attention mechanisms have been introduced into image fusion tasks. In [25], a fusion strategy based on CBAM was put forward by Li et al. This fusion strategy considered the significance of each spatial position and each channel. Similarly, Wang et al. [16] also constructed the spatial attention and channel attention models in the fusion layers. Unlike the Nestfuse method in [25], L1-norm and mean operation were combined with the attention models in [16] to generate a weight map for the salient features.

The deep learning-based fusion methods provide a competitive fusion performance. However, some drawbacks still need to be addressed. First, image fusion not only has the advantages of crucial information integration but also can boost high-level vision tasks. However, the previously published fusion methods [22], [45], [47], [48], [49] solely focus on the visual quality and statistical metrics of fused images, ignoring the application of image fusion in high-level vision tasks. Different from these methods, our method not only achieves superior performance in the image fusion task, but also contributes to improving the detection performance for subsequent high-level vision tasks, which is demonstrated by the extended application to salient object detection. Second, some fusion strategies are based on direct hand-crafted rules without any appropriate guidance when integrating the meaningful information of the source images, such as the addition operation in [21] and [22]. In contrast, we design a novel cross-modal image fusion strategy based on attention mechanism, which can integrate the meaningful information of the source images under the guidance of the attention mechanism. Although the attention mechanism-based methods in [45] and [49] can instruct the model to devote more focus on the important content, the long-range dependencies are ignored in the feature fusion process. Different from the methods in [45] and [49], our method employs the coordinate attention to design the feature-awareness fusion module and the feature-enhancement fusion module, which can concurrently capture channel and

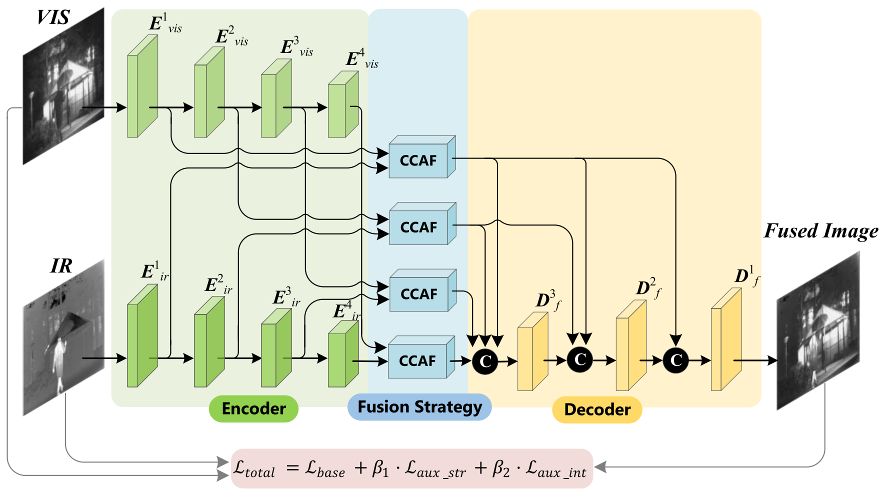


Fig. 2. Overall framework of the proposed network.

location importance and long-range dependencies. Third, the images with the same scene tend to have similar gray-level distributions, which can reflect the base information in image fusion, but the existing methods [48], [49] ignore this point. In contrast, we follow the fact that the gray-level distributions are similar in the same scene and develop a new loss function to impose similar gray-level distributions between the fused image and the source images by the KL divergence, thereby retaining the base information.

III. Proposed Method

In this section, we first discuss the motivation of our method. Then, we introduce the overall architecture of CCAFusion and illustrate the cross-modal coordinate attention fusion strategy in detail. Sequentially, we give the details of the designed multiple constrained loss function. Finally, we state the specific framework of CCAFusion for RGB-infrared image fusion.

A. Motivation

Infrared and visible image fusion aims to obtain one image with comprehensive information, which preserves rich texture characteristics in the visible image and thermal information in the infrared image. However, infrared and visible images are generated by different sensors with different imaging mechanisms, resulting in the domain discrepancy between the source images. The domain discrepancy is a great challenge for the infrared and visible image fusion task since it is difficult to balance the discrepancy of vital features between the source images. Many artifacts tend to emerge on the high-contrast edge regions of the fused image when the structure and intensity from source images can not coexist in harmony. Moreover, owing to the limited feature representation and the vital information loss during the fusion process, the fused results of the existing infrared and visible image fusion methods are also easily influenced by undesirable artifacts, which inevitably influence ultimate performance. By considering the above-mentioned limitations, we proposed a novel infrared and visible image fusion network, which aims to achieve adequate feature preservation, powerful feature representations, and the harmonious coexistence of vital features, thereby avoiding introducing undesirable artifacts to the fused images.

B. Overview

The proposed network follows a dual-stream model, which includes four encoder blocks (E^i_j $i \in \{1, 2, 3, 4\}$, $j \in \{ir, vis\}$), four fusion blocks, and three decoder blocks (D^i_f $i \in \{1, 2, 3\}$). Fig. 2 illustrates the overall framework of the proposed network. Specifically, the infrared image and visible image are first fed into the dual-stream feature extraction network, respectively. The backbone of each stream is an encoder-decoder network, which is based on the multiscale skip connection architecture [50], shown in Fig. 3. The encoder blocks aim to extract deep features at each level. Then, we convey the outputs from the encoder blocks to the cross-modal coordinate attention fusion blocks (CCAF), thereby achieving cross-modal fusion at each level. Successively, these cross-modal fused features are fed into the decoder blocks based on the skip connection architecture, and the final fused image is gradually reconstructed from the high-level information to the low-level information.

The following provides the detailed architecture of the encoder-decoder network. The encoder-decoder network is inspired by Densefuse [21]. Different from the architecture of Densefuse which obtains the reconstructed image through the dense connection-based network, we adopt a multiscale skip connection-based encoder-decoder network [50] to make full use of the multiscale features in the infrared image and the visible image, shown in Fig. 3. The encoder in each stream consists of four blocks, where each block includes two Padding layers, two convolution layers, and two ReLU layers. The kernel sizes in the first convolution layer and the second convolution layer are 3×3 and 1×1, respectively. The extracted features at different scales are integrated by the multiscale skip connection architecture and the concatenation operation. The results are then fed into three decoder blocks, respectively. Similar to the encoder part, each block of the decoder contains

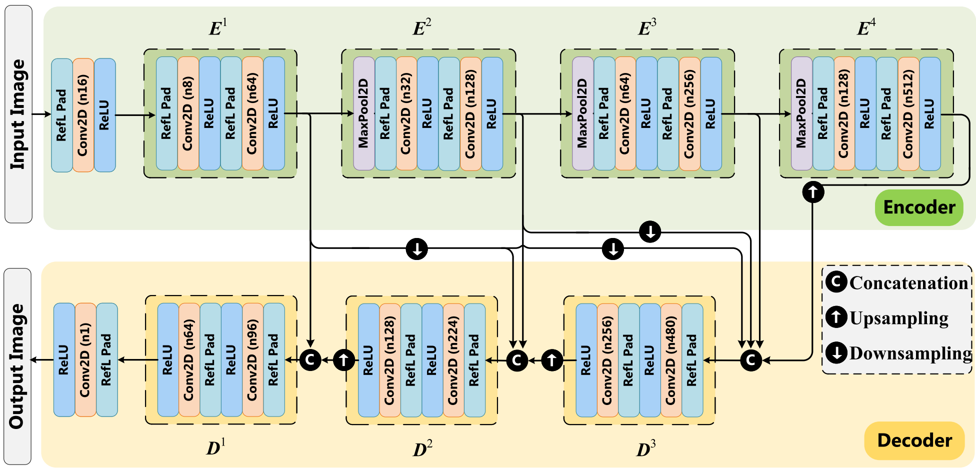


Fig. 3. Detailed architecture of the multiscale encoder-decoder network.

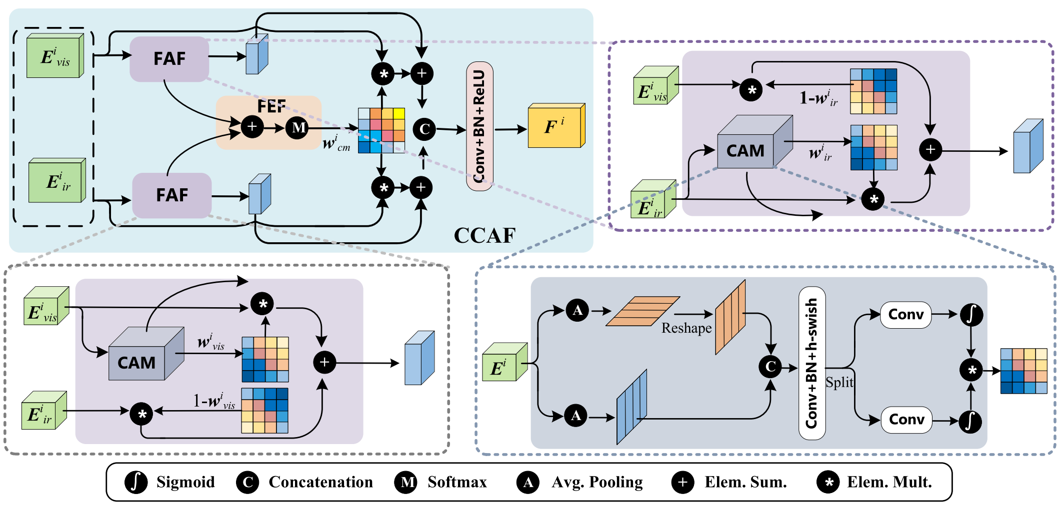


Fig. 4. Detailed architecture of the proposed fusion strategy.

two Padding layers, two convolution layers, and two ReLU layers. The channel numbers for the convolutional layers are reported in Fig. 3. The output image is generated by reducing the number of channels to one.

C. Fusion Strategy

The visible image contains rich textural features. The infrared image manifests thermal radiation features. These two types of features are complementary in complex environments, such as background clutter, occlusion, and light change. Considering that different spatial information and channels play different roles, we should focus on useful information when integrating these complementary features. Therefore, a novel attention mechanism is introduced into the fusion strategy to distinguish the importances of features.

The attention mechanism has been extensively studied and applied in a variety of fields, which can instruct the model to devote more focus on the important content [51], [52]. The most well-known and widely used approach is squeeze-and-excitation (SE) attention [53]. However, the significance of location information is disregarded in the SE module, which solely takes information between channels into account. CBAM takes both channel information and location information into account, which was employed for image fusion in [25]. However, this attention mechanism suffers the lack of long-range dependencies. Our fusion strategy is developed based on coordinate attention [28], which can concurrently capture channel and location importance and long-range dependencies. Fig. 4 shows the specific structure of our cross-modal fusion strategy, which includes two key parts: the feature-awareness fusion module (FAF) and the feature-enhancement fusion module (FEF).

We employ FAF to integrate complementary features, including the textural features in the visible image and thermal radiation features in the infrared image. First, the i-th feature E^i_{vis} generated by the visible stream and the i-th feature E^i_{ir} generated by the infrared stream are fed to CCAF. We employ the coordinate attention module (CAM) to obtain the weight matrices w^i_{vis} and w^i_{ir}, reflecting the importance of the visible features and the infrared features, respectively. The specific structure of CAM is shown in the bottom right part of Fig. 4. Given the i-th input feature E^i of size $C \times H \times W$, we employ pooling kernels $(H, 1)$ and $(1, W)$ to encode information of different channels along the horizontal coordinate and vertical

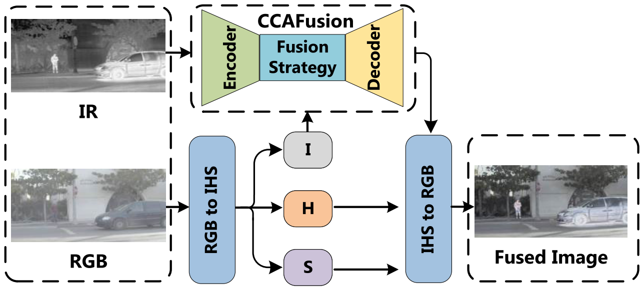


Fig. 5. The specific implementation for the RGB and infrared image fusion.

coordinate, respectively. The process of the c-th channel at height h is defined as follows:

$$y_{i\_c}^{h}(h) = \frac{1}{W}\sum_{0\le l<W} E^{i\_c}(h,l). \tag{1}$$

Similarly, the process of the c-th channel at width w is defined as follows:

$$y_{i\_c}^{w}(w) = \frac{1}{H}\sum_{0\le k<H} E^{i\_c}(k,w). \tag{2}$$

The above two processes yield a pair of direction-aware feature maps along two spatial directions, which allow the attention block to capture long-range dependencies along one spatial direction and preserve precise positional information along the other spatial direction, thereby helping the networks accurately locate the position of the interest objects. The intuitive effect of CAM is detailed in Section IV-F.3.

To produce the aggregated feature maps of reshaped $y_{i\_c}^{h}(h)$ and $y_{i\_c}^{w}(w)$, we concatenate them and reduce the number of channels to C/r by the 1×1 convolution operation. r is the reduction radio and set as 32 [28]. The formula is as follows:

$$\mathbf{f} = \delta\left(F_1\left(\left[y_{i\_c}^{h}(h), y_{i\_c}^{w}(w)\right]\right)\right), \tag{3}$$

where $[\cdot,\cdot]$ is the concatenation operation. F_1 denotes the 1×1 convolutional transformation function. δ is the h-swish activation function [54]. The output feature map $\mathbf{f} \in \mathbb{R}^{C/r\times(H+W)}$ is then decomposed into separate tensors $\mathbf{f}^h \in \mathbb{R}^{C/r\times H}$ and $\mathbf{f}^w \in \mathbb{R}^{C/r\times W}$. To restore the number of channels to C, we use two 1×1 convolutional transformation functions F_1^h and F_1^w. The weight matrix w^i is the multiplied output of F_1^h and F_1^w, which can be written as:

$$w^i = \sigma\left(F_1^h\left(\mathbf{f}^h\right)\right)\times\sigma\left(F_1^w\left(\mathbf{f}^w\right)\right), \tag{4}$$

where σ is the sigmoid function. We can obtain the weight matrices w_{vis}^i and w_{ir}^i by feeding E_{vis}^i and E_{ir}^i into CAM. Then, these two weight matrices are multiplied with the input features to focus on the important features and suppress the unnecessary ones for the fusion task, described as:

$$F_{ir}^i = E_{ir}^i * w_{ir}^i + E_{vis}^i * (1-w_{ir}^i), \tag{5}$$
$$F_{vis}^i = E_{vis}^i * w_{vis}^i + E_{ir}^i * (1-w_{vis}^i). \tag{6}$$

F_{ir}^i and F_{vis}^i are the FAF results. In addition, some textural features are also reflected in the infrared image, and some thermal radiation features are also reflected in the visible image. Consequently, FEF is applied to promote these complementary features. We first aggregate the weight matrices w_{vis}^i and w_{ir}^i. The aggregated result is then fed to the softmax function and the cross-modal weight matrix w_{cm}^i is obtained, expressed as:

$$w_{cm}^i = softmax(w_{vis}^i + w_{ir}^i). \tag{7}$$

Based on the cross-modal weight matrix w_{cm}^i, the enhanced features F_{IR}^i and F_{VIS}^i are obtained by adding the FAF results and the FEF results, expressed as:

$$F_{IR}^i = F_{ir}^i + E_{ir}^i * w_{cm}^i, \tag{8}$$
$$F_{VIS}^i = F_{vis}^i + E_{vis}^i * w_{cm}^i. \tag{9}$$

The enhanced features generated by the visible branch and the infrared branch are further concatenated and fed to the convolutional layer to obtain the cross-modal fused feature F^i. The formula is presented as follows:

$$F^i = Conv([F_{VIS}^i,\ F_{IR}^i]). \tag{10}$$

D. Loss Function

Aiming at the problem of lacking desirable ground-truth data in image fusion, we design the loss function of the network training from multiple aspects to preserve more meaningful information in the infrared and visible image fusion task.

On one hand, we consider that the gray-level distributions are similar in the same scene. In view of this, we combine the gray-level distribution and the Kullback-Leibler (KL) divergence loss to design the base loss $\mathcal{L}_{base}$, which is formulated as:

$$\mathcal{L}_{base} = \left(\left(KL\left(S\left(p_f\right)||S\left(q_{ir}\right)\right)^2 + KL\left(S\left(p_f\right)||S\left(q_{vis}\right)\right)^2\right)/2\right)^{1/2}, \tag{11}$$

where q_{ir} and q_{vis} denote the gray-level distributions of the infrared image and the visible image, respectively. p_f stands for the gray-level distributions of the fused image. $S(\cdot)$ indicates the softmax function, which is utilized to rescale the elements. The KL divergence is applied to impose similar gray-level distributions between the fused image and the source images, so as to maintain the base information in the source images.

On the other hand, we consider that the detailed information and the thermal radiation information should be fully ensured in the fused images. The details of fused images can be expressed as the maximum aggregate of infrared and visible image textures, thereby maintaining rich detailed information. The element-wise maximum selection denotes an optimal intensity distribution for fused images, thereby highlighting the thermal radiation information. Based on these points, we construct an auxiliary structure loss $\mathcal{L}_{aux\_str}$ and an auxiliary intensity loss $\mathcal{L}_{aux\_int}$, which can be presented as:

$$\mathcal{L}_{aux\_str} = \left\|\left|\nabla I_f\right| - \max\left(\left|\nabla I_{ir}\right|, \left|\nabla I_{vis}\right|\right)\right\|_1, \tag{12}$$
$$\mathcal{L}_{aux\_int} = \left\|I_f - \max\left(I_{ir}, I_{vis}\right)\right\|_1, \tag{13}$$

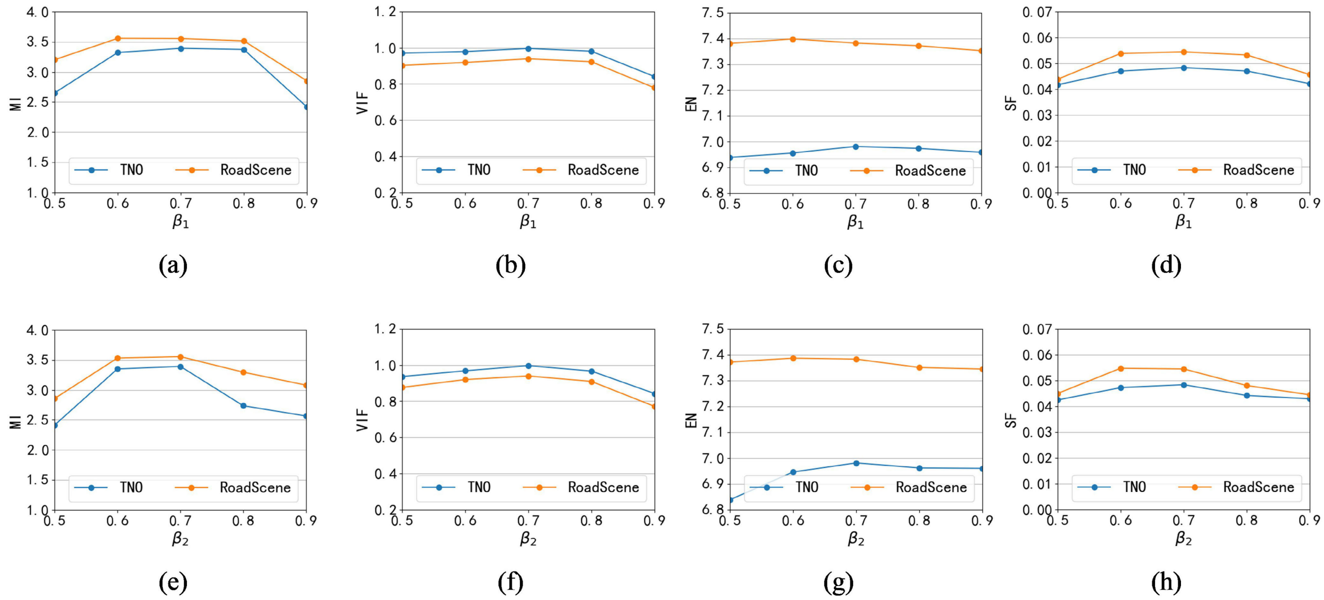


Fig. 6. The effect of hyperparameters in the multiple constrained loss function. First row: evaluations on MI, VIF, EN, and SF with different values of β_1. Second row: evaluations on MI, VIF, EN, and SF with different values of β_2.

where I_f, I_{ir} and I_{vis} are the fused image, the infrared image and the visible image, respectively. ∇ denotes the gradient operator.

Finally, the loss function of CCAFusion is composed of two parts: the base loss and the auxiliary loss. The auxiliary loss further consists of the auxiliary structure loss and the auxiliary intensity loss. The multiple constrained total loss function $\mathcal{L}_{total}$ is expressed as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{base} + \beta_1 \cdot \mathcal{L}_{aux\_str} + \beta_2 \cdot \mathcal{L}_{aux\_int}, \tag{14}$$

where β_1 and β_2 are the weights, which are utilized to strike a balance between different items. The specific adjustments about the values of β_1 and β_2 are demonstrated in Section IV-B.1.

E. RGB-Infrared Fusion

In the previous content, the stated fusion process is for the grayscale visible and infrared images. For the RGB and infrared image fusion, we can first transform the RGB image to the IHS color space. I represents the intensity channel. H and S stand for the hue channel and the saturation channel, respectively. Since the main spatial information is distributed in the I channel, we can employ CCAFusion to merge the infrared image and the I channel generated by the RGB image. The final fused image can be obtained by transforming into the RGB color space after combining the fused I channel with the H and S channels. The corresponding architecture for the RGB and infrared image fusion is illustrated in Fig. 5.

IV. Experiments and Results

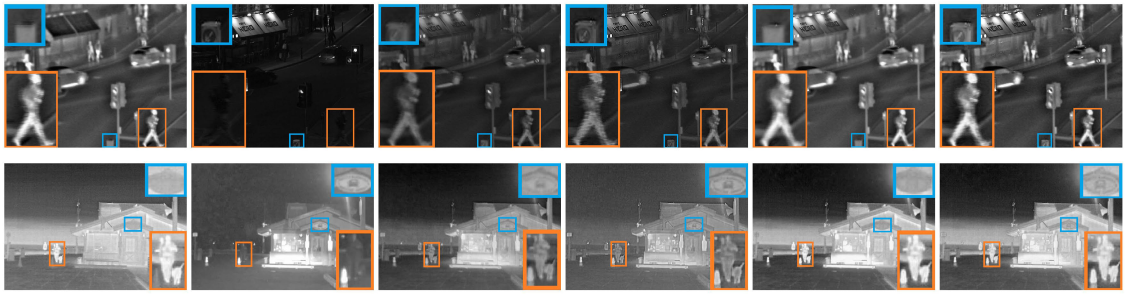
In this section, we first state the experimental settings, including datasets, evaluation metrics, compared methods, and implementation details. Then, we provide ablation experiments to verify the effectiveness of each component in our CCAFusion, and compare the proposed network with state-of-the-art image fusion methods. In addition, we conduct generalization experiments to demonstrate the generalizability of the proposed CCAFusion. Finally, we provide the extended application to salient object detection to illustrate the facilitation of our CCAFusion for the high-level vision task.

A. Experimental Settings

1) Datasets and Evaluation Metrics: We train the proposed network on the VTUAV dataset [55], which is the currently latest publicly-available multimodal dataset with visible-thermal. It contains 15 scenes across two cities, including the bridge, sea, road, park, and street. These various scenes cover different weather, including foggy days, windy days, and cloudy days. To form training images, we randomly select 10000 pairs from the VTUAV dataset. It should be noted that these training images are converted to grayscale and resized to 256×256.

We test the fusion performance of the proposed network on two public datasets, including 21 pairs of gray visible and infrared images collected from TNO [56] and 61 pairs of RGB and infrared images collected from RoadScene [27]. Note that the TNO dataset is pre-registered with several multiband camera systems and comprises multispectral (i.e., intensified visible, near-infrared, and long-wave infrared or thermal) images of different military and surveillance scenarios. RoadScene is an RGB-infrared aligned image dataset based on the FLIR video [57], and comprises image pairs with rich scenes, including roads, vehicles, and pedestrians.

To comprehensively assess the fusion quality, the fused images are evaluated from both qualitative assessment and quantitative metrics. The qualitative evaluation is based on human visual perception. However, subjective visual perception is frequently influenced by human factors. In view of this, we also adopt widely used quantitative metrics to assess



(a)IR (b)VIS (c) $\mathcal{L}_{base}$ (d)$\mathcal{L}_{base}$ &$\mathcal{L}_{aux\_str}$ (e)$\mathcal{L}_{base}$ &$\mathcal{L}_{aux\_int}$ (f) CCAFusion(ours)

Fig. 7. Ablation experiments for the multiple constrained loss function. From left to right: the infrared image, the visible image, and the fused images with the base loss $\mathcal{L}_{base}$, the base loss $\mathcal{L}_{base}$ and the auxiliary structure loss $\mathcal{L}_{aux\_str}$, the base loss $\mathcal{L}_{base}$ and the auxiliary intensity loss $\mathcal{L}_{aux\_int}$, and the multiple constrained loss function.

TABLE I
ABLATION EXPERIMENTS FOR THE MULTIPLE CONSTRAINED LOSS FUNCTION. RESULTS LABELED BY RED FONT REPRESENT THE BEST FUSION PERFORMANCE

Type			TNO				RoadScene			
$\mathcal{L}_{base}$	$\mathcal{L}_{aux\_str}$	$\mathcal{L}_{aux\_int}$	MI ↑	VIF ↑	SF ↑	EN ↑	MI ↑	VIF ↑	SF ↑	EN ↑
✓			1.849±0.39	0.680±0.18	0.039±0.01	6.513±0.40	2.356±0.54	0.626±0.10	0.050±0.01	6.776±0.22
✓	✓		1.957±0.40	0.690±0.17	0.042±0.02	6.827±0.42	2.539±0.53	0.639±0.11	0.051±0.01	7.247±0.22
✓		✓	2.413±0.57	0.801±0.28	0.028±0.01	6.906±0.34	2.971±0.39	0.778±0.13	0.032±0.01	7.330±0.17
✓	✓	✓	**3.393±0.60**	**0.997±0.28**	**0.048±0.01**	**6.981±0.27**	**3.554±0.43**	**0.939±0.14**	**0.054±0.01**	**7.383±0.20**

the fusion quality of different methods, including mutual information (MI) [58], visual information fidelity (VIF) [59], spatial frequency (SF) [60], and entropy (EN) [61]. MI can measure the amount of information that is transferred from the input images to the final fused image. The higher the value of MI, the more information the fused image contains. VIF can evaluate the information fidelity of the fused results. The bigger VIF is in better correspondence with human visual perception. SF can assess the richness of texture structure. The larger SF indicates that the fused result contains richer textures. EN can evaluate the amount of information remaining in the fused results. The larger EN means more information in the fused results.

2) Compared Methods and Implementation Details: To validate the fusion performance of CCAFusion, the proposed network is compared with state-of-the-art image fusion methods, including the traditional image fusion method and the deep learning-based image fusion methods. The traditional image fusion method is MDLatLRR [62]. The deep learning-based image fusion methods include DenseFuse [21], SDNet [43], PIAFusion [26], and U2Fusion [27]. For the compared methods, the default parameter settings are adopted.

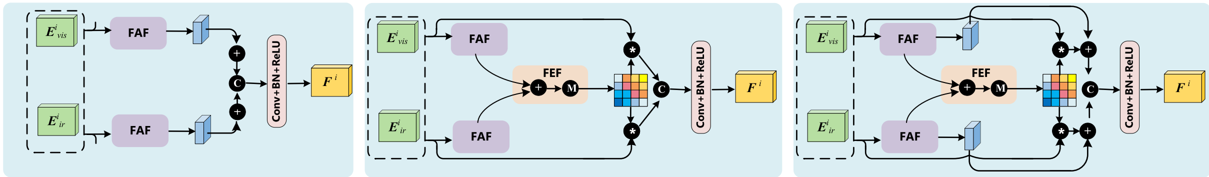
The experiment of MDLatLRR[1] is implemented on MATLAB R2020b with an AMD R7-4800H 2.9GHz CPU. The experiments of DenseFuse,[2] SDNet,[3] PIAFusion,[4] and U2Fusion[5] are based on TensorFlow with one Nvidia 2080Ti GPU. The proposed network is implemented in Pytorch and trained on one Nvidia 2080Ti GPU. We select the Adam optimizer with a learning rate of 0.0001. The batch size is set as 8, and the epoch is set as 10. It takes about 9h to train CCAFusion.

[1]https://github.com/hli1221/imagefusion_mdlatlrr
[2]https://github.com/hli1221/imagefusion_densefuse
[3]https://github.com/HaoZhang1018/SDNet
[4]https://github.com/Linfeng-Tang/PIAFusion
[5]https://github.com/hanna-xu/U2Fusion

B. Ablation Studies

1) Effect of Hyperparameters: In our CCAFusion, hyperparameters include β_1 and β_2 in the multiple constrained loss function. β_1 is applied to balance the base loss and the auxiliary structure loss. The base loss and the auxiliary intensity loss are balanced by β_2. We expect that the major information in the source images can be maintained in the fused image by the base loss. The auxiliary structure loss and the auxiliary intensity loss are of secondary importance, which are utilized to further ensure the texture features and the thermal radiation information in the fused images. Thus, we set the range of hyperparameters β_1 and β_2 from 0 to 1 to implement sensitivity analysis. We first fix the value of β_2 and test different β_1 to train the CCAFusion network. Second, we fix the value of β_1 and test different β_2 to train the CCAFusion network. Extensive experiments show that the best performance on the four evaluation metrics is achieved when $\beta_1 = 0.7$ and $\beta_2 = 0.7$, shown in Fig. 6.

2) Effect of Loss Function: To demonstrate the effectiveness of the multiple constrained loss function, we conduct an ablation study on the three terms of the designed loss function, including the base loss $\mathcal{L}_{base}$, the auxiliary structure loss $\mathcal{L}_{aux\_str}$, and the auxiliary intensity loss $\mathcal{L}_{aux\_int}$. Fig. 7 and Table I show the fused results with different loss functions. From these fused results, we can observe that the proposed network only ensures the main gray-level distribution when utilizing the base loss $\mathcal{L}_{base}$, and some thermal information and texture characteristics are lost (shown in Fig. 7(c)). With the base loss $\mathcal{L}_{base}$ and the auxiliary structure loss $\mathcal{L}_{aux\_str}$, although the details in the source images are transferred to



(a) w/o FEF (b) w/o FAF (c) CCAFusion (ours)

Fig. 8. Ablation architectures for fusion module. From left to right: structures without feature-enhancement fusion module (FEF), without the feature-awareness fusion module (FAF), and with the proposed fusion strategy.

TABLE II
ABLATION EXPERIMENTS FOR FUSION MODULE. RESULTS LABELED BY RED FONT REPRESENT THE BEST FUSION PERFORMANCE

Type	TNO				RoadScene			
	MI ↑	VIF ↑	SF ↑	EN ↑	MI ↑	VIF ↑	SF ↑	EN ↑
w/o FAF	2.526±0.66	0.894±0.20	0.042±0.01	7.054±0.24	3.012±0.46	0.837±0.12	0.044±0.01	**7.463±0.17**
w/o FEF	2.874±0.86	0.952±0.22	0.045±0.01	**7.086±0.28**	3.400±0.58	0.881±0.15	0.052±0.01	7.396±0.20
CCAFusion(ours)	**3.393±0.60**	**0.997±0.28**	**0.048±0.01**	6.981±0.27	**3.554±0.43**	**0.939±0.14**	**0.054±0.01**	7.383±0.20

TABLE III
ABLATION EXPERIMENTS FOR ATTENTION MECHANISM. RESULTS LABELED BY RED FONT REPRESENT THE BEST FUSION PERFORMANCE

Type	TNO				RoadScene			
	MI↑	VIF↑	SF↑	EN↑	MI↑	VIF↑	SF↑	EN↑
CCAFusion+SE	3.340±0.59	0.992±0.28	0.047±0.02	6.958±0.31	2.908±0.47	0.841±0.13	0.044±0.01	7.381±0.20
CCAFusion+CBAM	3.375±0.58	0.989±0.26	0.046±0.02	**7.012±0.25**	2.966±0.44	0.793±0.15	0.044±0.01	**7.404±0.22**
CCAFusion+CAM	**3.393±0.60**	**0.997±0.28**	**0.048±0.01**	6.981±0.27	**3.554±0.43**	**0.939±0.14**	**0.054±0.01**	7.383±0.20

the fused results, the thermal targets are not salient (shown in Fig. 7(d)). The thermal information is prominent under the base loss $\mathcal{L}_{base}$ and the auxiliary intensity loss $\mathcal{L}_{aux\_int}$. However, the detailed characteristics are lost (shown in Fig. 7(e)). When the multiple constrained loss function is adopted for image fusion, both the thermal information and the texture features are maintained in the fused results (shown in Fig. 7(f)). What is more, the multiple constrained loss function reaches larger average values in each metric (listed in Table I), which shows the effectiveness of the designed loss function.

3) Effect of Fusion Module: To evaluate the effectiveness of key components in our fusion strategy, we separately remove FAF and FEF of CCAFusion to explore their effect on the fusion performance. Fig. 8 displays the fusion network without the feature-enhancement fusion module (termed as w/o FEF), the fusion network without the feature-awareness fusion module (termed as w/o FAF), and the fusion network of CCAFusion. The experimental results of each fusion type are listed in Table II. We can see that CCAFusion reaches larger average values on three metrics, i.e., MI, VIF, and SF. The optimal results on MI indicate that CCAFusion transfers more information from the source images to the fused results. The best results in terms of VIF are in better correspondence with human visual perception. The best results in terms of SF indicate that the proposed method has a superior ability to preserve the detailed information. Although EN of CCFusion follows a narrow gap with the networks w/o FEF and w/o FAF, CCAFusion still has a considerable performance compared with state-of-the-art image fusion methods provided in Section IV-C and Section IV-D, indicating that the fused results of CCAFusion still contain abundant information.

4) Effect of Attention Mechanism: To evaluate the effectiveness of coordinate attention on our CCAFusion network in contrast to other attention mechanisms, we substitute CAM with commonly used attention methods, including SE attention and CBAM. The remaining aspects remain consistent with the original CCAFusion network. The quantitative results of image fusion using different attention mechanisms are shown in Table III. Benefiting from the ability of CAM to capture essential channel information, location information, and long-range dependencies, the CCAFusion network using CAM achieves the optimal values of MI, VIF, and SF, and the weak distance from the optimal value of EN, which indicates the effectiveness of CAM on our CCAFusion network.

C. Comparative Experiment

To assess the fusion performance of CCAFusion, the proposed network is compared with state-of-the-art image fusion methods on the TNO dataset, including a qualitative comparison and a quantitative comparison.

1) Qualitative Comparison: To intuitively display the performance differences of various image fusion methods, three pairs of fused images from the TNO dataset are selected for a qualitative comparison, shown in Fig. 9-11. The images from top left to bottom right in each figure are the infrared image, the visible image, and the fused images of DenseFuse-add, DenseFuse-L1, MDLatLRR, SDNet, PIAFusion, U2Fusion, and CCAFusion in sequence. To clearly demonstrate the fusion difference, we select local regions (i.e., the blue box and the orange box) in each fused image. The local regions are then enlarged and placed in the corner.

Fig. 9 shows that the fused results of DenseFuse-add, MDLatLRR, SDNet, and U2Fusion maintain thermal targets to some extent but suffer from the shortcoming of artifacts. For instance, undesirable halos or bright edges appear on the edges of the house and tree branches in MDLatLRR, while

TABLE IV
QUANTITATIVE RESULTS OF DIFFERENT METRICS ON THE TNO DATASET. THE VALUES CONTAIN THE AVERAGE, STANDARD DEVIATION, AND P-VALUE. THE RED RESULTS INDICATE THE BEST FUSION PERFORMANCE, AND THE BLUE RESULTS REPRESENT THE SUB-OPTIMAL FUSION PERFORMANCE

Method	MI↑		VIF↑		SF↑		EN ↑	
	avg±std	p-value	avg±std	p-value	avg±std	p-value	avg±std	p-value
DenseFuse-add	2.203±0.53	6.27E-08	0.773±0.20	6.24E-03	0.035±0.01	3.49E-03	6.681±0.45	1.60E-02
DenseFuse-L1	3.113±0.68	1.77E-01	**0.896±0.28**	2.64E-01	0.036±0.01	4.32E-03	**6.857±0.33**	2.05E-01
MDLatLRR	1.484±0.35	4.07E-15	0.743±0.13	7.22E-04	**0.064±0.03**	2.70E-02	6.741±0.52	4.63E-02
SDNet	2.008±0.68	3.51E-08	0.704±0.17	2.63E-04	0.045±0.01	4.56E-01	6.491±0.30	3.51E-06
PIAFusion	**3.246±0.59**	4.43E-01	0.867±0.11	6.38E-02	0.043±0.02	3.11E-01	6.665±0.49	1.47E-02
U2Fusion	1.804±0.42	5.12E-12	0.751±0.19	2.49E-03	0.044±0.01	3.99E-01	6.757±0.40	7.64E-02
CCAFusion(ours)	**3.393±0.60**	-	**0.997±0.28**	-	**0.048±0.01**	-	**6.981±0.27**	-

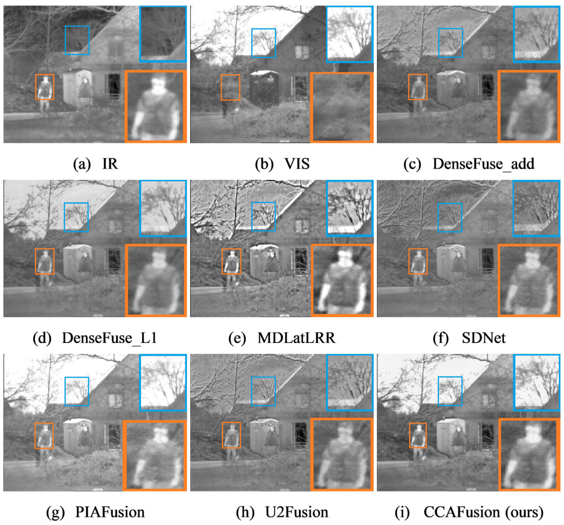


Fig. 9. Visual results generated by different image fusion methods (from top left to bottom right: the infrared image, the visible image, and the fused images of DenseFuse-add, DenseFuse-L1, MDLatLRR, SDNet, PIAFusion, U2Fusion, and CCAFusion) on $2\_men\_in\_front\_of\_house$ from the TNO dataset.

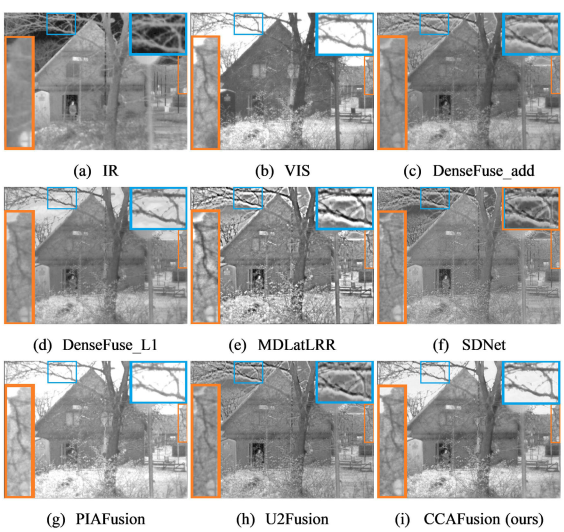


Fig. 10. Visual results generated by different image fusion methods (from top left to bottom right: the infrared image, the visible image, and the fused images of DenseFuse-add, DenseFuse-L1, MDLatLRR, SDNet, PIAFusion, U2Fusion, and CCAFusion) on $man\_in\_doorway$ from the TNO dataset.

this phenomenon also exists in the results in DenseFuse-add, SDNet, and U2Fusion (shown in the blue boxes). Although DenseFuse-L1 does not generate artifacts in the fused result, it lacks the contrast between the salient targets and the

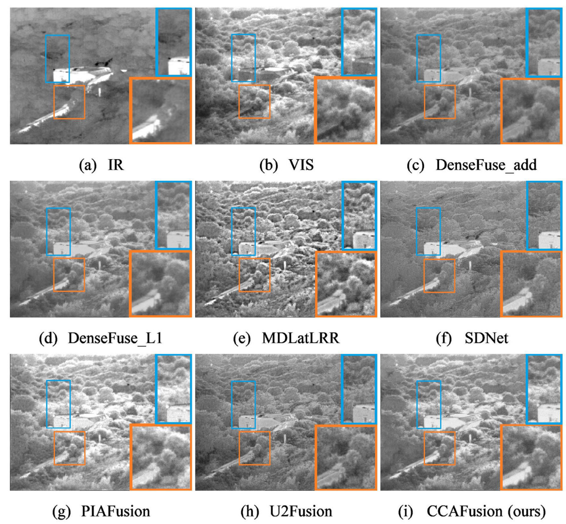


Fig. 11. Visual results generated by different image fusion methods (from top left to bottom right: the infrared image, the visible image, and the fused images of DenseFuse-add, DenseFuse-L1, MDLatLRR, SDNet, PIAFusion, U2Fusion, and CCAFusion) on $Bunker$ from the TNO dataset.

background region, resulting in inconsistency with the sky of the visible image, while PIAFusion maintains the background information but loses some textures of the tree branches. The texture of the tree branches in the fused image generated by CCAFusion is most consistent with the visible image. On the whole, CCAFusion can preserve the salient targets in the infrared image and details in the visible image without artifacts.

Fig. 10 and Fig. 11 show that the proposed CCAFusion can effectively merge the complementary information from the source images. Specifically, in Fig. 10, the sky in the other fused results is not consistent with the visible image or is contaminated by artifacts except for CCAFusion and PIA-Fusion. However, the thermal radiation targets are weakened by PIAFusion in the complex background region. In Fig. 11, the thermal radiation targets in the fused images generated by CCAFusion and SDNet are the most significant. However, the cloud in the background region of SDNet is not consistent with the visible image. Overall, our CCAFusion can not only effectively highlight the salient targets in the infrared images, but also has an advantage in maintaining the detailed texture in the visible images.

2) Quantitative Comparison: In addition to visual perception, we employ widely used quantitative metrics to further validate the fusion quality of CCAFusion. Fig. 12 and Table IV show the comparison results of various quantitative metrics on

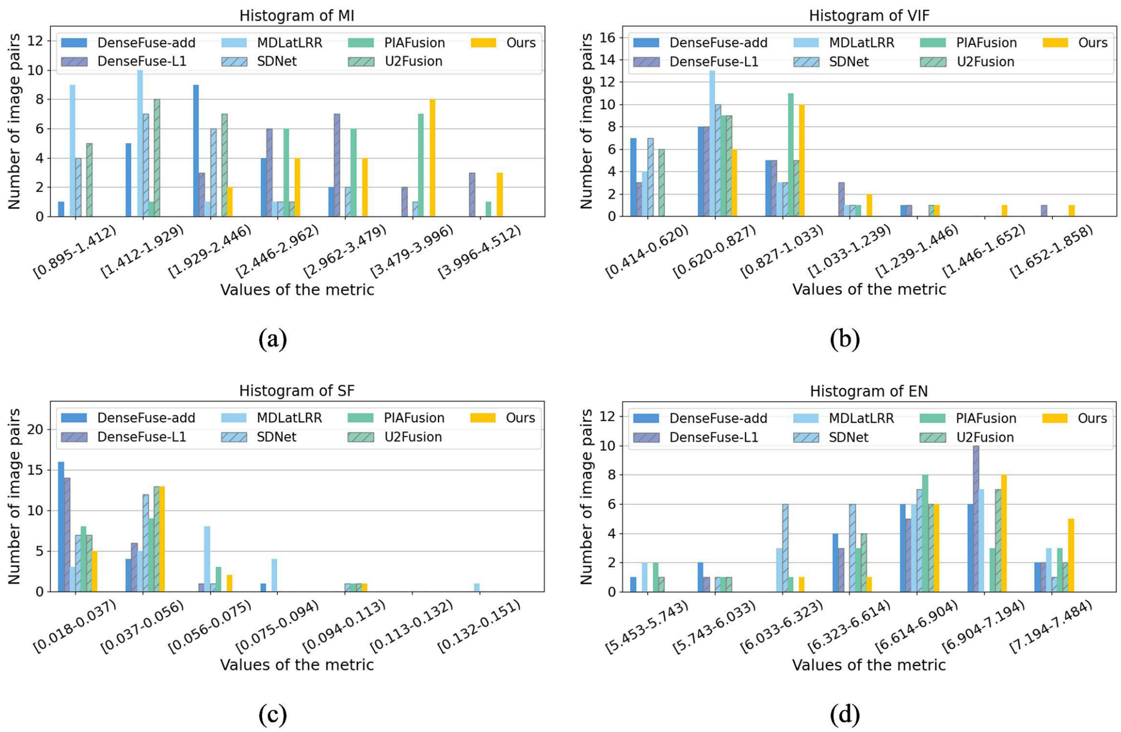


Fig. 12. Quantitative comparison of histogram distribution in different metrics (from top left to bottom right: MI, VIF, SF, and EN) on the TNO dataset. Horizontal coordinates indicate values of different metrics. Vertical coordinates represent the number of image pairs. Different bars stand for different image fusion methods.

21 image pairs from the TNO dataset. Specifically, Fig. 12 displays the comparisons of histogram distribution between existing image fusion methods and CCAFusion in various quantitative metrics. More right-skewed data mean greater metrics and better fusion performance. Table IV lists the average, standard deviation, and p-value on different metrics. Since the metrics with larger values indicate better fusion performance, the p-value p is based on a one-sided right-tail test, where $p = Pr\,(T \geq t \mid H_0)$. T denotes the test statistic from our method. t is the test statistic from compared methods. H_0 is the null hypothesis. Pr is the probability of T at least as extreme as t if H_0 is true. Moreover, values labeled by red font represent the best result, and values labeled by blue font indicate the sub-optimal result. Among the four metrics, note that CCAFusion has a remarkable advantage on three of them, namely MI, VIF, and EN. The higher MI indicates that the fused result generated by CCAFusion contains more information transferred from the source images to the fused image. The bigger VIF means that our fused image contains better information fidelity and is in better correspondence with human visual perception. In addition, CCAFusion provides the larger EN, which means more information in the fused results. In terms of SF, CCAFusion is only marginally lower than MDLatLRR. The reason is that artifacts and over-sharpening generated by MDLatLRR result in a higher SF.

D. Generalization Experiment

To demonstrate the generalizability of CCAFusion, we provide experiments on the RoadScene dataset. Since the visible images in the RoadScene dataset are RGB images, we employ the RGB-infrared fusion strategy described in Section III-E to achieve image fusion.

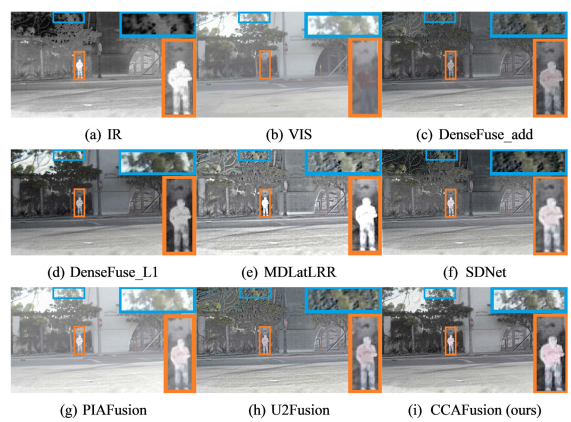


Fig. 13. Visual results generated by different image fusion methods (from top left to bottom right: the infrared image, the visible image, and the fused images of DenseFuse-add, DenseFuse-L1, MDLatLRR, SDNet, PIAFusion, U2Fusion, and CCAFusion) on $FLIR$_04598 from the RoadScene dataset.

1) Qualitative Comparison: Fig. 13-15 are three pairs of fused images from the RoadScene dataset, which show qualitative comparisons of various image fusion algorithms. To clearly observe the visual differences, the local areas are enlarged using the orange box and the blue box in each image. In Fig. 13-15, we can observe that the distorted information is introduced into the fused results of DenseFuse-add, MDLatLRR, SDNet, and U2Fusion, which is manifested by the distorted sky color (shown in Fig. 13 and Fig. 15) and the distorted streetlight color (shown in Fig. 14). Besides, the texture information and prominent targets in the source images

TABLE V

QUANTITATIVE RESULTS OF DIFFERENT METRICS ON THE ROADSCENE DATASET. THE VALUES CONTAIN THE AVERAGE, STANDARD DEVIATION, AND P-VALUE. THE RED RESULTS INDICATE THE BEST PERFORMANCE, AND THE BLUE RESULTS REPRESENT THE SUB-OPTIMAL PERFORMANCE

Method	MI↑		VIF↑		SF↑		EN ↑	
	avg±std	p-value	avg±std	p-value	avg±std	p-value	avg±std	p-value
DenseFuse-add	2.759±0.51	1.10E-15	0.724±0.10	1.51E-16	0.041±0.01	1.04E-09	7.140±0.24	2.33E-08
DenseFuse-L1	3.154±0.56	2.43E-05	**0.779±0.16**	4.44E-08	0.041±0.01	5.81E-09	7.223±0.24	1.45E-04
MDLatLRR	2.231±0.41	7.01E-19	0.755±0.10	1.47E-13	**0.084±0.03**	3.01E-13	**7.358±0.21**	5.08E-01
SDNet	**3.206±0.57**	2.42E-04	0.772±0.15	3.75E-09	0.052±0.01	2.69E-01	7.215±0.25	9.08E-05
PIAFusion	3.126±0.53	3.74E-06	0.776±0.15	9.88E-09	0.042±0.01	1.28E-08	7.021±0.21	1.21E-16
U2Fusion	2.702±0.53	1.35E-16	0.683±0.10	9.90E-21	0.042±0.01	3.06E-08	6.892±0.30	7.07E-19
CCAFusion(ours)	**3.554±0.43**	-	**0.939±0.14**	-	**0.054±0.01**	-	**7.383±0.20**	-

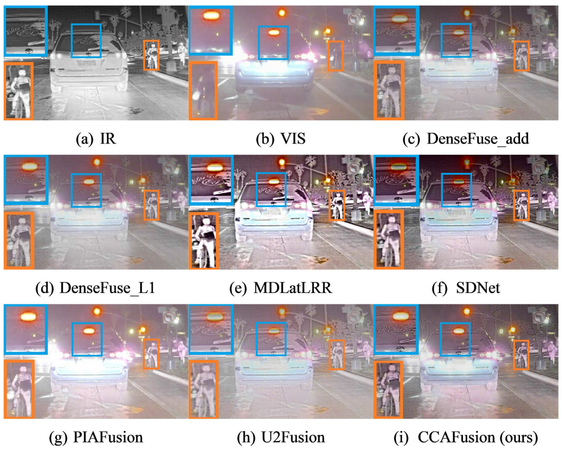


Fig. 14. Visual results generated by different image fusion methods (from top left to bottom right: the infrared image, the visible image, and the fused images of DenseFuse-add, DenseFuse-L1, MDLatLRR, SDNet, PIAFusion, U2Fusion, and CCAFusion) on $FLIR$_09624 from the RoadScene dataset.

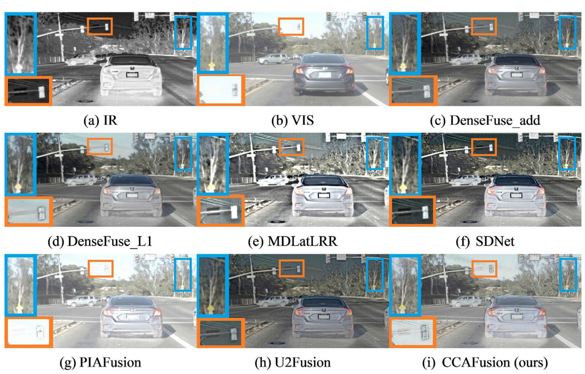


Fig. 15. Visual results generated by different image fusion methods (from top left to bottom right: the infrared image, the visible image, and the fused images of DenseFuse-add, DenseFuse-L1, MDLatLRR, SDNet, PIAFusion, U2Fusion, and CCAFusion) on $FLIR$_05164 from the RoadScene dataset.

are weakened in the fused images generated by DenseFuse-L1 and PIAFusion (shown in Fig. 13-15). It is worth mentioning that CCAFusion effectively preserves the texture details and salient targets in the source images.

2) Quantitative Comparison: In this section, the quantitative evaluations are conducted. The results of various quantitative metrics on 61 image pairs from the RoadScene dataset are shown in Fig. 16 and Table V. Similar to the quantitative comparison on the TNO dataset, histogram distributions of various quantitative metrics are shown in Fig. 16. More right-skewed data indicate higher metrics and better fusion capability. Table V reports the average, standard deviation, and p-value on different metrics. The best result is highlighted in red, and the second result is highlighted in blue. It can be seen that CCAFusion has the best values on MI, VIF, and EN. Such results indicate that the fused images generated by CCAFusion can not only maintain abundant information but also have satisfactory visual quality. The advantage of SF is not particularly pronounced in our CCAFusion due to the artifacts and over-sharpening in MDLatLRR.

TABLE VI

AVERAGE RUNNING TIME OF DIFFERENT METHODS ON THE TNO DATASET AND THE ROADSCENE DATASET (UNIT: SECOND)

Method	TNO	RoadScene
DenseFuse-add	0.442±0.43	0.325±0.15
DenseFuse-L1	0.851±0.34	0.549±0.16
MDLatLRR	146.694±50.67	70.851±17.57
SDNet	0.352±0.32	0.257±0.10
PIAFusion	1.502±1.05	1.004±0.37
U2Fusion	2.137±0.89	1.179±0.40
CCAFusion(ours)	0.183±0.07	0.080±0.03

E. Efficiency Comparison

To compare the processing speed of the proposed method with other methods, we record the average running time of different methods on the TNO dataset and RoadScene dataset, and the comparison results are reported in Table VI. Note that the traditional method (MDLatLRR) is implemented on AMD R7-4800H 2.90GHz CPU, and the other deep learning-based methods are conducted on Nvidia 2080Ti GPU. The average running time reflects the processing speed of the model. The standard deviation of running time reflects the robustness of the model. In Table VI, we can see that the deep learning-based methods are significantly faster than the traditional method, benefiting from the GPU acceleration. Compared with other deep learning-based methods, our method has competitive operational efficiency and comparable robustness. To this end, our method can be competently deployed as a preprocessing module for high-level vision tasks.

F. Application to Salient Object Detection

Salient object detection (SOD) is one of the common high-level vision tasks, which aims to estimate the conspicuous objects or regions in digital images. To further demonstrate the effectiveness of the proposed fusion method, we also feed the source images and fused results into the salient object detection method [63], respectively. VT821 [64], VT1000 [65], and VT5000 [66] are selected as the testing datasets, which are three public SOD datasets about infrared and visible image pairs.

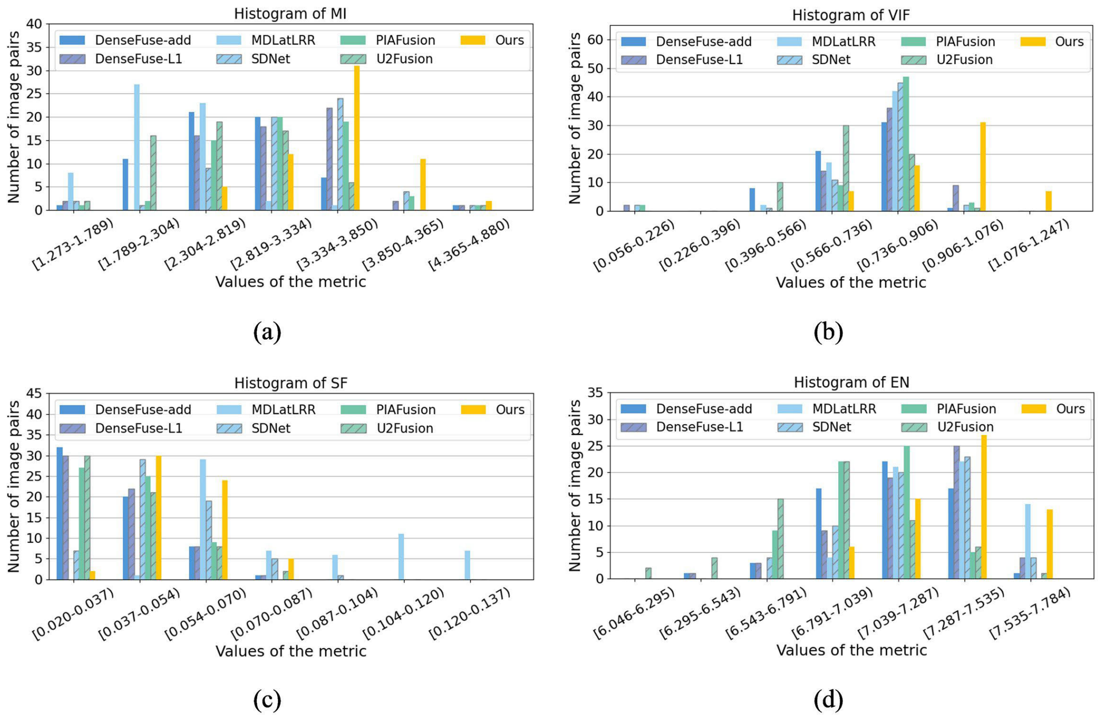


Fig. 16. Quantitative comparison of histogram distribution in different metrics (from top left to bottom right: MI, VIF, SF, and EN) on the RoadScene dataset. Horizontal coordinates indicate values of different metrics. Vertical coordinates represent the number of image pairs. Different bars stand for different image fusion methods.

TABLE VII

PERFORMANCE COMPARISON OF VISIBLE IMAGES, INFRARED IMAGES, AND FUSED IMAGES ON THREE TESTING DATASETS. THE RED RESULTS INDICATE THE BEST PERFORMANCE, AND THE BLUE RESULTS REPRESENT THE SUB-OPTIMAL PERFORMANCE

Type	VT821					VT1000					VT5000				
	Fm↑	MAE↓	Sm↑	P↑	R↑	Fm↑	MAE↓	Sm↑	P↑	R↑	Fm↑	MAE↓	Sm↑	P↑	R↑
VIS	0.754	0.060	0.837	0.754	0.839	0.861	0.031	0.903	0.859	0.926	0.768	0.052	0.834	0.775	0.852
IR	0.629	0.086	0.719	0.662	0.670	0.716	0.082	0.785	0.728	0.784	0.582	0.102	0.709	0.595	0.692
DenseFuse-add	0.747	0.063	0.831	0.746	0.846	0.850	0.035	0.895	0.842	0.928	0.761	0.059	0.833	0.757	0.877
DenseFuse-L1	0.748	0.060	0.838	0.746	0.854	0.868	0.029	0.909	0.863	0.936	0.773	0.051	0.843	0.771	0.875
MDLatLRR	0.761	**0.056**	0.843	0.761	**0.862**	0.865	0.029	0.907	0.859	0.936	0.776	0.050	**0.848**	0.771	**0.886**
SDNet	0.755	0.058	0.841	0.751	0.849	0.864	0.030	0.909	0.857	**0.944**	0.777	0.050	0.846	0.775	0.878
PIAFusion	0.748	0.061	0.838	0.745	0.847	0.868	0.029	**0.911**	0.862	0.943	0.780	0.050	0.844	**0.781**	0.874
U2Fusion	0.758	0.058	0.839	0.759	0.848	0.863	0.031	0.906	0.857	0.937	0.773	0.052	0.843	0.769	0.881
CCAFusion(ours)	**0.765**	0.057	**0.845**	**0.765**	**0.862**	**0.872**	**0.027**	**0.911**	**0.866**	0.939	**0.782**	**0.048**	0.847	**0.781**	0.878

1) Qualitative Comparison: We provide some typical examples in Fig. 17 to illustrate the advantages of our CCA-Fusion in facilitating salient object detection. It can be seen that the visible images cannot accentuate all the conspicuous objects because of the background limitation, which leads to incomplete regions in the detection results. The detection results based on the infrared images are far from the ground truth (GT) when the background is blended with the salient object. Compared with the detection results based on a single modality, the fused images play a positive role in salient object detection. It is worth mentioning that our fused images obtain the most consistent result with the ground truth since the powerful capability of integrating the thermal information and the texture information.

2) Quantitative Comparison: We further adopt quantitative metrics to evaluate the performance of salient object detection, shown in Table VII. P and R are the average precision and average recall, respectively. F-measure (Fm) denotes the weighted harmonic mean of the precision and recall. S-measure (Sm) evaluates the similarity of spatial structure between the ground truth and its predicted map. The higher values of P, R, Fm, and Sm, the better the detection performance. Mean Absolute Error (MAE) measures the average pixel-level relative error between the ground truth and its predicted map. The lower values of MAE, the better the detection performance. Table VII shows the performance comparison of the visible image, the infrared image, and fused images generated by our method and other methods. It can be seen that the fused images achieve better performance compared with a single modality, indicating the positive role of image fusion on salient object detection. Besides, the fused image of DenseFuse-add shows poorer performance on salient object detection owing to the direct addition strategy in the fusion layer. In contrast, the fused image of our CCAFusion achieves obvious advantages on most metrics, which demonstrates the effectiveness of the proposed method.

In general, the fused images of our CCAFusion not only perform well in the infrared and visible image fusion task but also contribute to improving the performance of salient object detection.

TABLE VIII
PERFORMANCE COMPARISON OF FUSED IMAGES GENERATED BY DIFFERENT ATTENTION MECHANISMS ON THREE TESTING DATASETS. THE RED RESULTS INDICATE THE BEST PERFORMANCE

Type	VT821					VT1000					VT5000				
	Fm↑	MAE↓	Sm↑	P↑	R ↑	Fm↑	MAE↓	Sm↑	P↑	R↑	Fm↑	MAE↓	Sm↑	P↑	R↑
CCAFusion+SE	0.753	0.059	0.840	0.749	0.848	0.864	0.031	0.906	0.856	0.942	0.776	0.051	0.844	0.773	0.876
CCAFusion+CBAM	0.755	0.058	0.843	0.749	0.850	0.864	0.031	0.907	0.857	**0.943**	0.776	0.050	0.845	0.774	**0.878**
CCAFusion+CAM	**0.765**	**0.057**	**0.845**	**0.765**	**0.862**	**0.872**	**0.027**	**0.911**	**0.866**	0.939	**0.782**	**0.048**	**0.847**	**0.781**	**0.878**

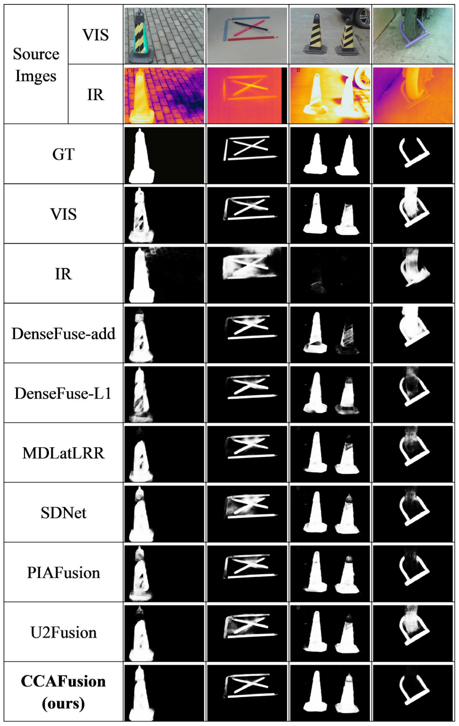


Fig. 17. Qualitative comparison of the predicted salient maps using different methods.

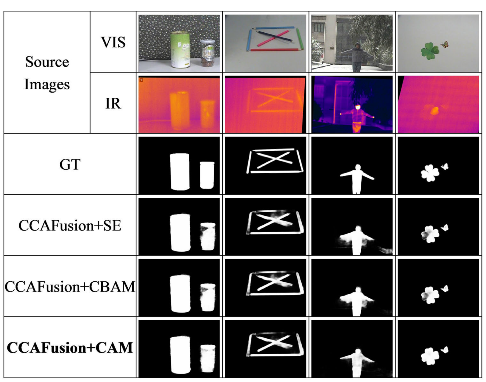


Fig. 18. Qualitative comparison of the predicted salient maps using different attention mechanisms.

3) Attention Mechanism Comparison: To further compare the performance of coordinate attention with other attention methods on our CCAFusion network, we replace CAM with other attention methods, including the widely adopted SE attention and CBAM. For other settings, we follow the original CCAFusion network. Qualitative and quantitative results using different attention mechanisms are shown in Fig. 18 and Table VIII, respectively. In Fig. 18, it can be seen that our CCAFusion network with CAM can help better in detecting the salient objects than SE attention and CBAM. The advantage of coordinate attention on CCAFusion is also reflected in Table VIII. Our CCAFusion network with CAM outperforms other attention methods on most metrics. We argue that the advantage benefits from essential channel information, location information, and long-range dependencies. However, the significance of location information is disregarded in the SE module, which solely takes information between channels into account. CBAM takes both channel information and location information into account, but it suffers the lack of long-range dependencies. CAM can concurrently capture channel and location importance and long-range dependencies. Therefore, we adopt CAM in our CCAFusion to obtain superior performance.

V. CONCLUSION

In this paper, we propose an effective cross-modal coordinate attention network for infrared and visible image fusion, named CCAFusion. Specifically, we design the cross-modal coordinate attention fusion strategy to integrate complementary features. To utilize the multi-level information, the designed fusion strategy is applied at different scales. Furthermore, we define a multiple constrained loss function consisting of the base loss and the auxiliary loss to adjust the gray-level distribution and ensure the harmonious coexistence of structure and intensity in fused images. Extensively comparative experiments reveal the advantage of CCAFusion over state-of-the-art image fusion methods in terms of qualitative assessment and quantitative metrics. Moreover, the extended experiments on salient object detection demonstrate that our CCAFusion can facilitate the performance of the high-level vision task.

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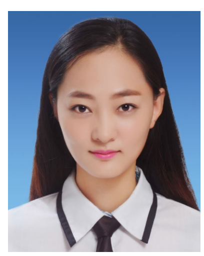

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